Evaluation Guide

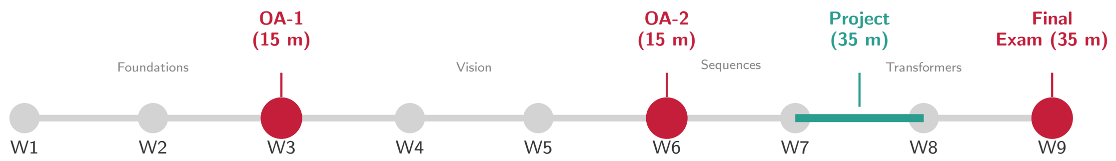
Scaler School of Technology | Deep Learning I: Neural Networks | Semester 4 | 100 Marks

Evaluation Overview

This document describes all four evaluation components of the course: two online assessments, a final group project, and a comprehensive final exam. Read this carefully so you know what to expect, when, and how to prepare.

#	Component	Format	When	Marks
1	Online Assessment 1	SCQ + Multi-MCQ + Descriptive	End of Week 3	15
2	Online Assessment 2	SCQ + Multi-MCQ + Descriptive	End of Week 6	15
3	Final Project	Group Project + Presentation + Viva	Weeks 7–8	35
4	Final Online Exam	SCQ + Multi-MCQ + Match + Code + Descriptive	Week 9	35
			Total	**100**

Assessment Timeline



General Policies

- Online assessments and the final exam have **negative marking** in the single-correct MCQ section (see details per component)
- All online assessments are **proctored** with browser lock-down
- **Late project submissions:** −5% per day, max 3 days
- **Plagiarism** results in zero credit for the component

Online Assessment 1 — 15 Marks

Schedule & Format

When: End of Week 3 **Duration:** 60 minutes

Mode: Online, proctored **Batches:** 2 (Group A & Group B)

Total: 15 marks

OA-1 Syllabus: Weeks 1–3

Week	Topics
1	Representation learning, UAT, perceptron, activations, MLPs, forward pass, XOR
2	Chain rule, computational graphs, backprop, loss functions, SGD, momentum, Adam, LR schedules
3	Overfitting, L2/dropout, early stopping, batch norm, Xavier/He init, vanishing/exploding gradients

Online Assessment 2 — 15 Marks

Schedule & Format

When: End of Week 6 **Duration:** 60 minutes

Mode: Online, proctored **Batches:** 2 (Group A & Group B)

Total: 15 marks

OA-2 Syllabus: Weeks 4–6

Week	Topics
4	Convolution operation, padding, stride, output size formula, pooling, receptive fields, CNN architectures (LeNet, AlexNet, VGG, GoogLeNet, ResNet), 1×1 convolutions
5	Transfer learning (feature extraction vs. fine-tuning), pre-trained models, object detection (YOLO, Faster R-CNN), vanilla RNN, hidden state dynamics, BPTT, vanishing gradients in RNNs
6	LSTM cell (forget/input/output gates), cell state updates, GRU, bidirectional & stacked RNNs, encoder-decoder, information bottleneck, attention (Bahdanau, Luong)

Final Project — 35 Marks

Schedule

Assigned: Week 4 **Presentations:** Weeks 7–8 **Group size:** 8 students (random)

Two Tasks Per Group

Task 1 (Section 1): An end-to-end applied project — build something real, train it, evaluate it. Builds **confidence** and hands-on skills.

Task 2 (Section 2): A focused experimental task designed to build **theoretical and conceptual foundations** — understand *why* things work, not just *how* to make them run.

Marks Breakdown

Component	Scope	Marks
Project & Presentation — Technical (problem understanding, implementation, experiments, analysis, report)	Group	∼21
Presentation Skills (delivery, communication, visual quality, demo)	Group	∼4
Individual Viva (project understanding + theory, Weeks 1–7)	Individual	∼11
Total		**35**

Deliverables (Due Day Before Presentation)

1. **Code Notebook (Task 1)** — .ipynb with saved outputs, must run top-to-bottom
2. **Code Notebook (Task 2)** — .ipynb with saved outputs, focused on controlled experiments
3. **Project Report** — PDF, 5–6 pages covering both tasks
4. **Slide Deck** — PDF or Google Slides, max 15 slides, every member presents at least 1 slide

Viva Format

- **One person** chosen randomly, announced **one day before**
- Lasts **8–10 minutes**: project + theory questions
- Their score becomes the viva score for the **entire group**
- Everyone must prepare — you don't know who will be called

Important Notes

- The viva carries ~**11 out of 35 marks** (30%) — one member's performance determines the group's viva score.
- **Vote-out policy:** If a member is not contributing, the group can vote them out (majority vote + evidence). The voted-out member must complete the same topic as an individual project.
- Code must **run without errors** on Google Colab.
- Late submissions: −5% per day, maximum 3 days late.
- Plagiarism (code or report) results in **zero credit** for the entire project.
- See the **Final Project Guide** for the complete list of topics (26 applied + 26 experimental) and detailed rubric.

Final Online Exam — 35 Marks

Schedule & Format

When: Week 9 **Duration:** 120 minutes

Mode: Online, proctored **Scope:** Comprehensive (Weeks 1–7)

The final exam includes single-correct MCQs, multiple-correct MCQs, match-the-following, code completion, and descriptive questions covering Weeks 1–7.

How to Prepare

For Online Assessments (OA-1 & OA-2)

- Review lecture slides and in-class notebooks for all sessions in the syllabus range.
- Practice numerical calculations: output sizes, parameter counts, loss computations.
- Understand the *why* behind each technique — the MCQs test conceptual understanding, not rote memorization.
- Re-do the hands-on exercises from each session's Colab notebook.
- There is negative marking in the assessments, so be careful when answering.

For the Final Project

- Start early — you have from Week 4 to Week 7/8. Don't leave it to the last week.
- Every team member should be able to explain any part of the project — the viva tests individual understanding.
- Include proper evaluation: accuracy alone is not enough. Use confusion matrices, loss curves, ablation studies.
- Practice your presentation — 12 minutes goes fast. Rehearse at least twice as a team.

For the Final Exam

- This is cumulative (Weeks 1–7). Start revision early — don't cram.
- Focus on understanding connections: Why do transformers outperform RNNs? How does attention solve the bottleneck?
- The exam has more depth than OA-1/OA-2 — expect derivations, architecture design, and code completion.

Quick Reference

	OA-1	OA-2	Project	Final Exam
Marks	15	15	35	35
When	End of Week 3	End of Week 6	Weeks 7–8	Week 9
Duration	60 min	60 min	—	120 min
Syllabus	Weeks 1–3	Weeks 4–6	Weeks 1–7	Weeks 1–7
Mode	Online	Online	In-person	Online

Course: Deep Learning I: Neural Networks **Semester:** 4 **Instructor:** Priyansh Saxena